Word-Level Plagiarism Checker for Short Documents

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DECLARATION

The Mini Project Report entitled "**Word-Level Plagiarism Checker for Short Documents"** is a record of Bonafide work of **Lankey Tejasri -2520090017, Dikshita Siripireddy - 2520090007** submitted in partial fulfillment for the award of B. Tech in Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the mini project based report entitled "**Word-Level Plagiarism Checker for Short Documents"** is a bonafide work done and submitted by **Lankey Tejasri -2520090017, Dikshita Siripireddy - 2520090007.** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science and Information Technology, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

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1.ABSTRACT

The **Word-Level Plagiarism Checker for Short Documents** is a plagiarism detection system designed to identify copied and lightly paraphrased content in short text documents such as essays, abstracts, reports, and assignment answers. The project compares an input document with a reference corpus at the word level using custom-built data structures instead of prebuilt library implementations. It detects exact matches, reordered sentences, and reworded passages by analyzing word sequences and similarity patterns. The system also identifies the original source of each plagiarized section and highlights the matching text for easy review, making the results clear and easy to understand. Optimized for fast processing, it delivers results suitable for interactive use by instructors, editors, publishers, and students. By combining accuracy, efficiency, detailed source attribution, and user-friendly output, the project provides a reliable solution for plagiarism detection while demonstrating the practical implementation of fundamental data structures and algorithms in a real-world application.

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2. INTRODUCTION

The rapid growth of digital content has made it easier to create, share, and reuse information. In academic and professional environments, this has also increased the possibility of plagiarism, where content is copied or modified from existing sources without appropriate acknowledgment. Detecting such content is important for maintaining originality and ensuring the credibility of written work. However, manually comparing documents can be time-consuming, particularly when multiple sources need to be examined.

The **Word-Level Plagiarism Checker for Short Documents** is developed to provide an efficient approach for identifying similarities in short documents such as essays, abstracts, reports, and assignment answers. The project focuses on comparing an input document with reference documents and examining similarities at the word level. It considers different forms of similarity, including exact matches, reordered sentences, and reworded passages.

The system further helps users understand the detected plagiarism by identifying the corresponding source and highlighting matching sections. This makes the results easier to review and interpret. The project demonstrates the application of data structures and algorithms to a practical text-processing problem while focusing on efficient and understandable results.

3. LITERATURE SURVEY

1. Turnitin

Turnitin is a widely used plagiarism detection system designed primarily for academic institutions. It compares submitted documents against a large collection of web pages, academic publications, and previously submitted student papers. The system generates similarity reports that show matching content and its corresponding sources. It is mainly used by universities and instructors to evaluate the originality of academic submissions.

https://www.turnitin.com/

2. Grammarly Plagiarism Checker

Grammarly provides plagiarism detection as part of its writing assistance platform. It checks written content against online sources and academic databases to identify potentially copied material. Along with plagiarism checking, it provides grammar, spelling, and writing suggestions. This combination makes it useful for students and writers who want to check originality while improving their written content.

https://www.grammarly.com/plagiarism-checker/

3. Scribbr Plagiarism Checker

Scribbr is an academic-focused plagiarism detection platform intended mainly for students and researchers. It checks documents for copied and paraphrased content and provides detailed similarity reports. The system is particularly designed for academic papers, essays, theses, and dissertations. Its reports help users identify matching sources and review potentially problematic sections before submission.

https://www.scribbr.com/plagiarism-checker/

4. Quetext

Quetext is an online plagiarism checker that uses its search technology to identify similarities between submitted text and existing sources. It is designed for students, educators, writers, and content creators. The system provides highlighted results and source information to make detected similarities easier to understand. It focuses on identifying both direct copying and similarities that may result from modified wording.

https://www.quetext.com/

5. Copyleaks

Copyleaks is a plagiarism detection platform designed for academic, professional, and organizational use. It can analyze documents for similarities across online and academic sources and provides detailed reports. The platform also supports detection across multiple languages and includes additional AI-content analysis capabilities. Its broader source coverage makes it suitable for organizations that require more comprehensive content verification.

https://copyleaks.com/

6. DupliChecker

DupliChecker is an online plagiarism checking tool that provides a simple way to scan text for duplicated content. It is commonly used for quick checks of essays, articles, assignments, and other written material. The system compares submitted text with online content and reports possible matches. Its simple interface makes it suitable for users who need a basic and accessible plagiarism-checking solution.

https://www.duplichecker.com/

7. SmallSEOTools Plagiarism Checker

SmallSEOTools provides a web-based plagiarism checker intended for quick content verification. Users can enter or upload text and scan it for similarities with online sources. The tool is mainly useful for basic plagiarism checking and provides results showing potentially duplicated content. Its easy accessibility makes it useful for students, writers, and users who need an initial originality check.

https://smallseotools.com/plagiarism-checker/

4. Problem Statement and Objectives

Problem Statement: The difficulty of efficiently detecting copied, reordered, and reworded content in short documents and identifying their original sources.

Problem Description: Manual plagiarism checking of essays, reports, abstracts, and assignments is time-consuming and may not effectively identify modified or rearranged content. A system is needed to compare documents, detect similarities, identify sources, and highlight matching content.

Objectives:

- To develop a system for detecting copied and modified content in short documents.
- To reduce the time and effort required for manual plagiarism checking.
- To compare an input document with reference documents and identify similar content.
- To detect exact matches, reordered sentences, and reworded passages.
- To identify the original source of the matching content.
- To highlight similar or plagiarized sections for easy verification.
- To provide clear, efficient, and understandable plagiarism detection results.
- To apply data structures and algorithms to a real-world text comparison problem.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB of free disk space

Software Requirements:

- Operating System: Windows 10/11
- Programming Language: Java
- IDE: Visual Studio Code or Eclipse

6. IMPLEMENTATION

Module 1: User Registration

File Name: `register.html`

Code Snippet:

```
<form id="registerForm">

    <label for="registerName">Full Name</label>
    <input
        type="text"
        id="registerName"
        placeholder="Enter your name"
        required
    >

    <label for="registerEmail">Email Address</label>
    <input
        type="email"
        id="registerEmail"
        placeholder="Enter your email"
        required
    >

    <label for="registerPassword">Password</label>
    <input
        type="password"
        id="registerPassword"
        placeholder="Create a password"
        minlength="6"
        required
    >

    <label for="confirmPassword">Confirm Password</label>
    <input
        type="password"
        id="confirmPassword"
        placeholder="Confirm your password"
        minlength="6"
        required
    >

    <button type="submit" class="login-button">
        Create Account
    </button>

</form>
```

Module 2: Plagiarism Detection

File Name: `js/plagiarism.js`

```
function analyzeDocument(inputText) {
    const inputWords = tokenize(inputText);
    let similarity = 0;

    REFERENCE_CORPUS.forEach(reference => {
        const referenceWords = tokenize(reference.text);
        const jaccard = calculateJaccard(
            inputWords, referenceWords
        );
        similarity = Math.max(similarity, jaccard * 100);
    });

    return {
        similarity: Math.round(similarity),
        original: Math.round(100 - similarity),
        wordsAnalyzed: inputWords.length
    };
}
```

Module 3: Document Upload and Plagiarism Checking.

File Name: `checker.html`

```
<input type="file" id="fileInput" accept=".txt,.pdf">

<textarea id="documentText"
    placeholder="Paste your document text here...">
</textarea>

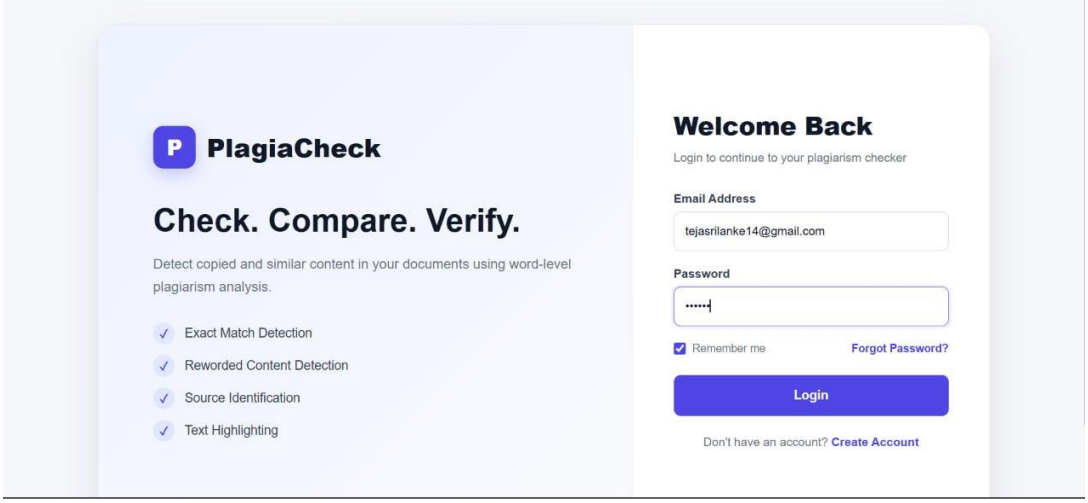
<button type="button" id="checkButton">
    Check Plagiarism
</button>

<script>
checkButton.addEventListener("click", async function() {
    const response = await fetch("http://127.0.0.1:8000/check", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({
            text: documentText.value
        })
    });
    const result = await response.json();
    localStorage.setItem(
        "latestPlagiarismResult",
        JSON.stringify(result)
    );
    window.location.href = "results.html";
});
</script>
```

7. RESULTS & DISCUSSION

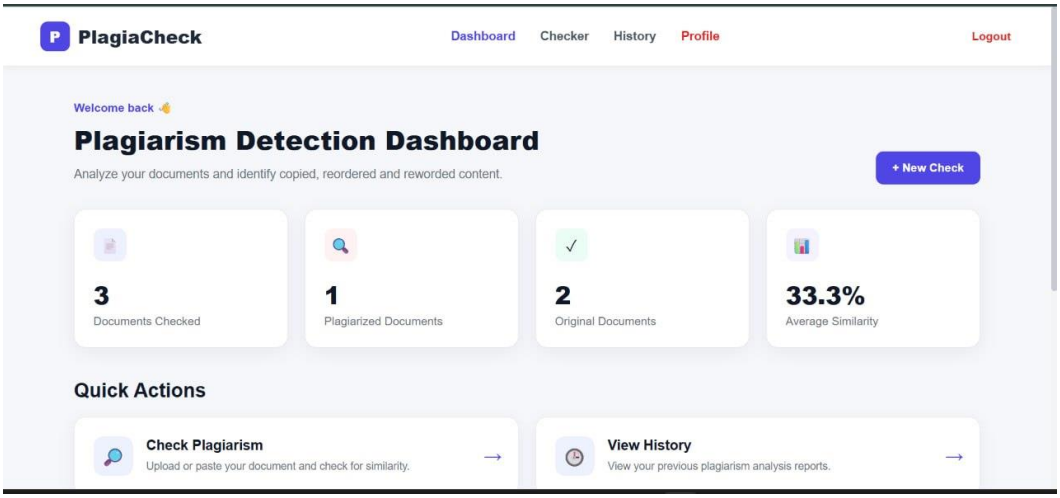
Login Page

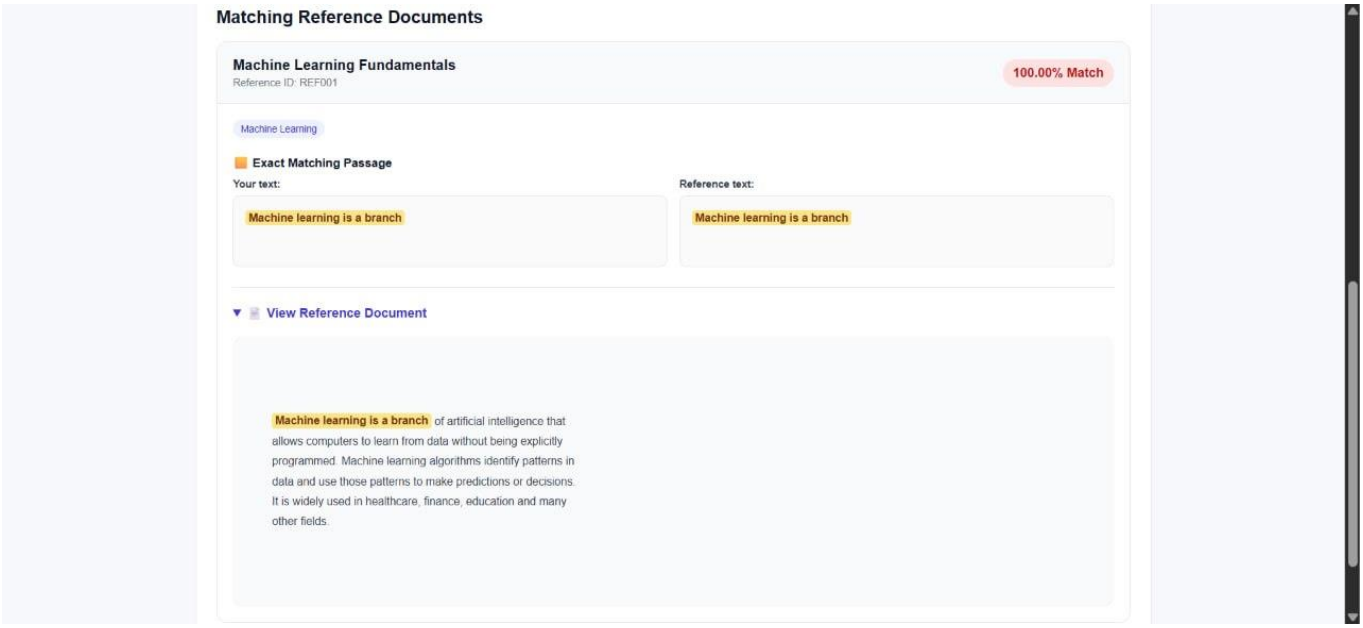
The login page allows registered users to enter their email address and password to access PlagiaCheck. It provides a simple interface with a login button and a create-account option.



Plagiarism Detection Dashboard

The dashboard displays the number of documents checked, plagiarized documents, original documents, and average similarity percentage. It also provides quick access to plagiarism checking and previous analysis reports.





The system identifies the source document containing matching text, highlights the identical passages and displays the corresponding similarity percentage.

8. CONCLUSION & FUTURE SCOPE

Conclusion

The Word-Level Plagiarism Checker for Short Documents aims to identify copied and similar content in short documents such as essays, abstracts, reports, and assignments. The project uses text preprocessing, word-sequence generation, and similarity comparison to detect matching content between an input document and reference documents. It also aims to identify the source of matching text and present the results in an understandable format. The project demonstrates the practical application of Data Structures and Algorithms in solving a real-world problem.

Future Scope

The project can be enhanced in the following ways:

1. Short-Term Enhancements

- Develop a user-friendly web interface for uploading and checking documents.
- Support multiple file formats, such as TXT, PDF, and DOCX.
- Improve the highlighting and display of matching text.

2. Medium-Term Enhancements

- Implement advanced algorithms to detect reordered sentences and lightly paraphrased content.
- Expand the reference database to compare documents against a larger collection.
- Improve processing speed and accuracy for multiple documents.

3. Long-Term Enhancements

- Integrate Natural Language Processing (NLP) and machine learning techniques for semantic similarity detection.
- Develop a web-based platform for students, teachers, and researchers.
- Explore multilingual plagiarism detection to support documents written in different languages.

9. REFERENCES

1. Turnitin https://www.turnitin.com/
2. Grammarly Plagiarism Checker https://www.grammarly.com/plagiarism-checker
3. Quetext https://www.quetext.com/plagiarism-checker
4. Copyleaks https://copyleaks.com/
5. Scribbr Plagiarism Checker https://www.scribbr.com/plagiarism-checker/
6. Plagiarism Checker X https://plagiarismcheckerx.com/
7. DupliChecker https://www.duplichecker.com/
8. Plagiarism Detector https://plagiarismdetector.net/
9. SmallSEOTools Plagiarism Checker https://smallseotools.com/plagiarism-checker/
10. Prepostseo Plagiarism Checker https://www.prepostseo.com/plagiarism-checker